

Learning experience – Business Analytics

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First of all I would like to address my learning curve that I experienced in programming and data analysis. As already discussed in our coaching session, I feel like the result was a good thing for me in this class. And I actually learned the basics of Studio R which I now needed for further data analysis.

However, this project was one of the most challenging yet rewarding learning experiences of the semester. At the beginning of the course, I had limited experience with complex coding in R and data analysis. Many of the concepts, such as data cleaning, handling missing values, detecting duplicates, creating visualizations, and running statistical analyses, were completely new to me. So, I initially felt overwhelmed by the amount of new knowledge and the technical skills required.

But the Netflix Data Analysis project allowed me to apply the concepts from the lectures in a practical setting. Personally, I feel like this is what helped me to understand coding a bit better. It was not a fictional scenario anymore but rather a real case with real data that I could grasp. One of the first challenges was understanding the structure of the dataset and identifying potential data quality issues. Through the cleaning process, I learned how important it is to inspect data before performing any analysis. And how much work it actually can be to clean a raw dataset like this. I became more familiar with functions such as `filter()`, `mutate()`, `count()`, and techniques for handling missing values and duplicates. At first, even simple coding tasks required a lot of trial and error, (and perhaps one or two crying sessions) but over time I became more confident in reading and modifying R code.

A significant part of my learning curve involved data visualization. This is when all our hard work suddenly materialized into something visual. Using `ggplot2`, I saw how raw data transformed into meaningful charts that communicated the insights we wanted effectively. Creating visualizations such as the comparison of Movies versus TV Shows, the top producing countries, genre distributions, and the growth of Netflix content over time helped me understand how graphical representations can support data-driven decision-making. Later, we also experimented with `gganimate`, which introduced us to more advanced visualization techniques and demonstrated how animations can make trends easier to understand. This was actually fun to watch and try out. But in the future we will be more careful to not overstimulate my colleagues' laptop.

Another important aspect of the project was the statistical analysis. After the coaching session we decided to also work with linear regression. I learned how relationships between variables can be modeled and interpreted. Before this project, statistical methods often seemed abstract and theoretical. Although I understood them better than algebra math, applying them to a real dataset helped me understand their practical relevance even more and gave me greater confidence in interpreting results.

Beyond coding and statistics, I gained valuable experience in reporting and presenting analytical findings. Creating a professional presentation in Quarto was initially difficult because I had never worked with this tool before. But again, it proved to be very rewarding. Learning how to structure slides, insert executable R code chunks, customize the appearance with CSS, and render the final presentation taught me how to communicate technical results effectively to an audience. This part of the project showed me that data analysis is not only about obtaining results but also about presenting them in a clear and engaging way.

One of the most important lessons I learned throughout this project was the value of my own persistence. Many problems could not be solved immediately, and debugging code often required patience and careful investigation. However, each challenge I overcame contributed to my understanding and improved my confidence in working with data and programming.

Looking back, I can clearly see my progress from someone who was unfamiliar with R programming to someone who can clean data, perform analyses, create visualizations, and develop professional reports and presentations. While I still have much to learn, this project provided a strong foundation in data analytics and significantly improved my technical skills. Which I can now use in my future job. Most importantly, it showed me that coding is not only a technical skill but also a powerful tool for solving real-world problems and linking them to a business base while generating meaningful insights from data.